Technical Documentation

Diesel Engine
12 V 956 TB31
Generating Set

Description and
Operating Instructions
M010410/00E

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M010410/00E

Description and Operating Instructions

12V 956 TB31

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MTU Friedrichshafen GmbH
Abteilung MSD
88040 Friedrichshafen
GERMANY

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LEGAL UNITS OF MEASUREMENT WITH MTU-F

	Unit	SI-Symbol	further permissible symbols (W.E.) hitherto customary symbols etc.
Length, Area, Volume	Length	m	Basic unit W.E.: μm, mm, km the hitherto "Micron" and μ for 0.001 mm not permitted anymore
	Area [A]	m^2	W.E.: mm^2, a (Ar) = 100 m^2, ha Abbreviations with q, for ex. qm only permitted till 12. 1974
	Volume	m^3	W.E.: l (litre), mm^3, cm^3, dm^3 Abbreviations cbm, ccm, cmm only permitted till 12. 1974
Angle	Plane angle	rad * *Radiant*	1 rad = 1m (arc) : 1 m (radius) W.E.: degree (°) minute (') second (")
Mass	Mass	kg *Kilogramme*	Basic unit W.E.: mg, g, t hitherto kgs^2/m and kps^2/m Alteration of numerical value Weight statements further as hitherto used in g, kg, t when understood as mass
	Density	kg/m^3	W.E.: g/cm^3, kg/l hitherto: kgs^2/m^4 and kps^2/m^4 Alteration of numerical values
	Mass moment 2^{nd} grade *Moment of inertia*	kg m^2	hitherto: $kgms^2$ and $kpms^2$ Alteration of numerical values
Time	Time *Periodic time*	s *Second*	Basic unit W.E.: minute (min.), hour (h), day (d) with min., h, d no decimal prefixes do not use as span: week, month, year do not write: sec., sek., Sek, Min., Min, St, Std.
	Frequency	Hz *Hertz*	1 Hz = 1/s W.E.: kHz, MHz
	Velocity	m/s	W.E.: km/h
	Acceleration [a]	m/s^2	
	Angular velocity	rad/s	W.E.: °/s
	Angular acceleration	rad/s^2	
	Speed	-	rpm, 1/min, 2π rad/min, rev/min for example: n = 800 rpm, n = 500/min.
	Impulse (momentum)	Ns	hitherto kgs, kps Alteration of numerical values
	Torsional momentum *Twist*	Nsm	hitherto kgsm, kpsm Alteration of numerical values
	Specific fuel consumption	kg/J*	W.E.: g/kWh hitherto g/PSh Alteration of numerical values

Dr. No. 60 047e

Force, Energy, Power

Unit	SI-Symbol	further permissible symbols (W.E.) hitherto customary symbols etc.
Force [F]	N *Newton*	1 N = 1 kgm/s^2 W.E.: mN, kN, MN hitherto: g, kg, t and p, kp, Mp Alteration of numerical values 1 kp = 9.80665 N $\approx$ 10 N, 1 Mp $\approx$ 10 kN etc. for <u>forces of weight</u> caused by masses also N, mN, kN, MN are valid
Pressure *Especially of fluids*	Pa * *Pascal*	1 Pa = 1 N/m^2 W.E.: <u>bar</u> (Bar), mbar 1 bar = 10^5Pa = 10 N/cm^2 = 0.1 N/mm^2 = 1.01972 at hitherto: kg/cm^2, kp/cm^2, at, atm, ata, atü, mmWS, Torr, mm Hg (QS) Alteration of numerical values 1 kp/cm^2 = 1 at = 0.980665 bar 1 mmWS = 0.0981 mbar 1 mmHg (QS) = 1 Torr = 1.333 mbar if overpressure, differential pressure, underpressure or absolute pressure <u>differentiate by using the respective terms</u>
Mech. stress, strength, elasticity-, thrust-, *(slide)* module	N/m^2*	W.E.: N/mm^2 hitherto: kg/mm^2, kp/mm^2 Alteration of numerical values 1 kp/mm^2 = 9.81 N/mm^2 $\approx$ 10 N/mm^2 (1N/mm^2 = 0.102 kp/mm^2 $\approx$ 0.1 kp/mm^2) Example: 40 kp/mm^2 $\approx$ 392 N/mm^2
Energy Work Force	J *Joule*	1 J = 1 Nm = 1 Ws = 1 kgm^2/s^2 hitherto: kgm, kpm Alteration of numerical values 1 kpm $\approx$ 9.81 J
Quantity of heat		W.E.: kJ hitherto: cal, kcal Alteration of numerical values 1 kcal $\approx$ 4.187 kJ $\approx$ 1.163 Wh
Momentum of force, *torque bending moment*	Nm J*	hitherto: kgm, kpm Alteration of numerical values 1 kpm $\approx$ 9.81 Nm
Notch impact strength	J/m^2*	W.E.: J/mm^2, J/cm^2 hitherto: kgm/cm^2, kpm/cm^2 Alteration of numerical values
Power [P] Energy flow Heat flow	W* *Watt*	1 W = 1 J/s = 1 Nm/s W.E.: <u>kW</u>, kJ/h hitherto: PS, kgm/s, kpm/s Alteration of numerical values 1 PS = 75 kpm/s = 75·9.81 Nm/s = 0.736 kW

Viscometrical values

Dynamic viscosity	Pas	1 Pas = 1 Ns/m^2 = 1 kg/ms W.E.: mPas hitherto: P *(Poise)* 1 P = 0.0102 kps/m^2 cP *(Centipoise)* 1 cP = 1mPas
Kinematic viscosity	m^2/s*	W.E.: mm^2/s hitherto: St *(Stokes)* 1 St = 1 cm^2/s cSt *(Centistokes)* 1 cSt = 1mm^2/s

Temperature, Heat

Temperature Temp. interval	K *Kelvin*	<u>Basic unit</u> W.E.: ^{o}C ($0^{o}C$ ≙ 273.15 K; + 100 ^{o}C ≙ 373.15 K) with ^{o}C no decimal prefixes ^{o}K and grd not permitted any more after 1975
Specific heat capacity	J/kg K*	W.E.: kJ/kgK hitherto: kcal/kg ^{o}C Alteration of numerical values
Calorific value	J/kg* J/m^3*	W.E.: kJ/kg, kJ/m^3 hitherto: kcal/kg, $kcal/m^3$ Alteration of numerical values

Remarks: * To be avoided with MTU-F

[ ] means: Altered, new symbols according to DIN 1304

Italics imply: Additional explanation, prefer symbols underlined

PREFACE

This publication is intended to inform about the MTU Diesel engine and aid in its operation, servicing and amintenance.

The section Technical Data contains details of the engine output, weight, settings and operational values etc.

The section Description and Directions for Servicing contains, besides the description of the engine components, instructions for performing inspections, servicing and maintenance.

> As regards the inspection and maintenance operations which must be carried out at regular intervals, the time sequences of the maintenance schedule are valid.

> Directions for operations of greater extent, which go beyond normal inspection and maintenance, are contained in the Assembly Instructions.

In the section Operating Instruction hints are given for operating the engine and its supervision before, during and after operation.

The section Maintenance lists the maintenance operations and a maintenance schedule.

> The time schedule indicates at which intervals the maintenance operations have to be carried out.

The section Hints for Trouble Shooting names possible fault symptoms and gives hints on the cause and also the correction of the fault.

The section Appendix contains illustrations and drawings.

The model designation and specifications given by the manufacturers of the ancillaries and equipment mounted to the engine conform to the engine equipment at the time of issue of this publication.

These particulars may in no case be used for ordering spare parts.

For the ordering of spare parts is authorized exclusively the spare parts catalogue.

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